

ISyE 3770 HW1

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Problem 2.1.13

Using the Region Numbers as outlined in class, we have:

- **Part (a)** A' : First, we identify A as the green circle. Since part (a) asks for the elements not in A , the answer is: $3 + 5 + 6 + 8$.
- **Part (b)** $A \cap B$: First, we find the elements that exist in both A and B . Therefore, we find the answer to be: $1 + 2$.
- **Part (c)** $(A \cap B) \cup C$: We begin by identifying the intersection of A and B , which we see to be $1 + 2$ as previously found. Now, we will find the elements that are in C and add them to the list. Finally, we have the following as our answer: $1+2+3+4+5$.
- **Part (d)** $(B \cup C)'$: We will start by identifying the union of B and C to be $1 + 2 + 3 + 4 + 5 + 6$. Now, we will find the answer not to be $7+8$.
- **Part (e)** $(A \cap B)' \cup C$: First, we will identify A intersection B or, the elements both A and B have in common. We find this to be $1 + 2$. Then, we will find answer not to be $3 + 4 + 5 + 6 + 7 + 8$. Then, we will find the answer union C to be $1+3+4+5+6+7+8$.

Problem 2.1.21

Using the definitions of A and B , we have the following:

- **Part (a)** A' : We begin by identifying A to be elements less than 72.5. Therefore, we find A' to be elements greater than or equal to 72.5. Symbolically, we have: $A' = \{x \mid x \geq 72.5\}$.
- **Part (b)** B' : We find B to be elements greater than 52.5. Therefore, we find B' to be elements less than or equal to 52.5. Symbolically, we have $B' = \{x \mid x \leq 52.5\}$.
- **Part (c)** $A \cap B$: The intersection of A and B include elements in both A and B . Therefore, we find such elements to be those between 52.5 and 72.5, exclusive. Symbolically, we have $A \cap B = \{x \mid 52.5 < x < 72.5\}$.
- **Part (d)** $A \cup B$: The union of A and B include elements that are in A or B (or both). Therefore, we find such elements to be those less than 72.5 and those greater 52.5. Symbolically, we have $A \cup B = \{x \mid x > 0\}$.

Problem 2.2.13

We must find samples containing at least four defective parts, so we must find two answers: samples containing 4 defective parts and samples containing 5 defective parts.

We recognize order does not matter, so we proceed using a combination.

For 4 defective parts, we know we must pick four of the five defective parts then we must pick six of the forty-five other parts. Therefore, we have

$$C_4^5 C_6^{45}$$

For 5 defective parts, we follow similar logic:

$$C_5^5 C_5^{45}$$

Therefore, we find our total number of samples to contain at least four defective parts to be $C_4^5 C_6^{45} + C_5^5 C_5^{45} = 40,725,300 + 1,221,759 = 41,947,059$.

Problem 2.3.1

- **Part (a)** $P(A)$: We know the probability of an event is the sum of the probabilities of the outcomes in the event. As such, we have:

$$P(A) = P(a, b, c) = P(a) + P(b) + P(c) = 0.1 + 0.1 + 0.2 = 0.4$$

- **Part (b)** $P(B)$: Following the previous question's logic, we have:

$$P(B) = P(c, d, e) = P(c) + P(d) + P(e) = 0.2 + 0.4 + 0.2 = 0.8$$

- **Part (c)** $P(A')$: We remember $P(A) = 1 - P(A')$ and so, we have our answer:

$$P(A') = 1 - P(A) = 1 - 0.4 = 0.6$$

- **Part (d)** $P(A \cup B)$: Since the union of A and B cover all events in the sample space, we arrive at our answer of $P(A \cup B) = 1$.
- **Part (e)** $P(A \cap B)$: There exists only one element in the intersection of A and B and so, we arrive at our answer of $P(A \cap B) = P(\{c\}) = 0.2$.

Problem 2.3.17

- **Part (a)** $P(A)$: The probability of A is the probability of a password, containing 8 characters, being composed of only letters. For the first character, we know there are 62 total choices of characters, but 52 that are a letter. We will use this logic 8 times, so we have:

$$P(A) = \left(\frac{52}{62}\right)^8 \approx 0.24485$$

- **Part (b)** $P(B)$: The probability of B is the probability of a password, containing 8 characters, being composed of only integers. Using similar logic as previously, we have:

$$P(B) = \left(\frac{10}{62}\right)^8 = 4.58 \times 10^{-7}$$

- **Part (c)** $P(\text{Password containing at least 1 integer})$: The probability of a password containing exactly 1 integer follows $P(A') = 1 - P(A)$, so we have $P(A') = 1 - 0.24485 = 0.75515$.
- **Part (d)** $P(\text{Password containing exactly 2 integers})$: We wish for two of the 8 positions in the password to be an integer. Therefore, we have $C_2^8 = 28$. For the two selected positions, there exist 10 possible integers, or 100 possible characters. For the remaining 6 positions, we have 52 options, resulting in 52^6 possibilities. Therefore, the number of passwords containing exactly two integers is:

$$28 * 100 * 52^6$$

Therefore, we arrive at our answer of $\frac{28 * 100 * 52^6}{62^8} = 0.2535$.

Problem 2.4.7

- **Part (a)** $P(A)$: Since $P(A)$ represents the probability of selecting a password beginning with a vowel, we know we C_1^5 represents choosing a vowel to start the password and, to fill the remaining 5 characters, there are 36^5 options. Therefore, we have the following:

$$P(A) = \frac{C_1^5 * 36^5}{36^6} = \frac{5}{36}$$

- **Part (b)** $P(B)$: Abstracting this problem, it is asking the probability of one of five characters being in a particular position of the password... which is the same as **part (a)**. As such, we have:

$$P(B) = \frac{C_1^5 * 36^5}{36^6} = \frac{5}{36}$$

- **Part (c)** $P(A \cap B)$: Using the intersection rule of probability, we have $P(A \cap B) = P(A) * P(B)$. We recognize events A and B are independent of each other - the last character being an even number does not influence the probability of the first being a vowel. As such, we have:

$$P(A \cap B) = P(A) * P(B) = \frac{5}{36} * \frac{5}{36} = \frac{25}{1296}$$

- **Part (d)** $P(A \cup B)$: Using the union rule of probability, we have $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. As such, we have the following:

$$P(A \cup B) = \frac{5}{36} + \frac{5}{36} - \frac{25}{1296} = 0.2584$$

Question 2: Poker

To find the number of combinations for choosing three of the same cards, we must first choose a rank, represented by C_1^{13} . Then, we must choose three of the four cards of the rank: C_3^4 . We must then select one of the remaining 12 ranks and choose one card: $C_1^{12}C_1^4$. Then, we must do this once more with the remaining 11 ranks: $C_1^{11}C_1^4$. Finally, we have:

$$C_1^{13} * C_3^4 * C_1^{12} * C_1^4 * C_1^{11} * C_1^4 =$$

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Question 3: Four Candidates

This question follows very similar to a repeated combination example. In this case, we have $n = 4$ and $r = 100$, or a selection of 100 items with repetition from a set of 4. As such, we have:

$$C_r^{n+r-1} = C_{100}^{4+100-1} =$$

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